

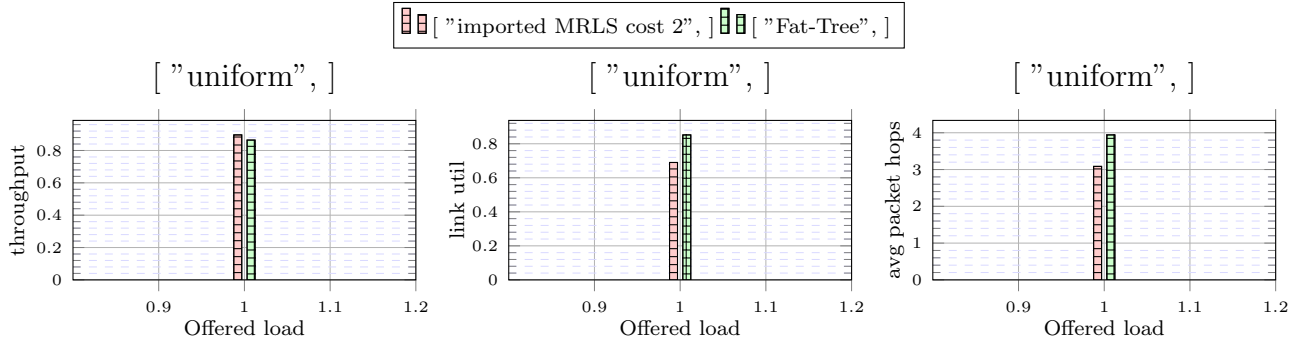


Figure 1: X: ["uniform",]

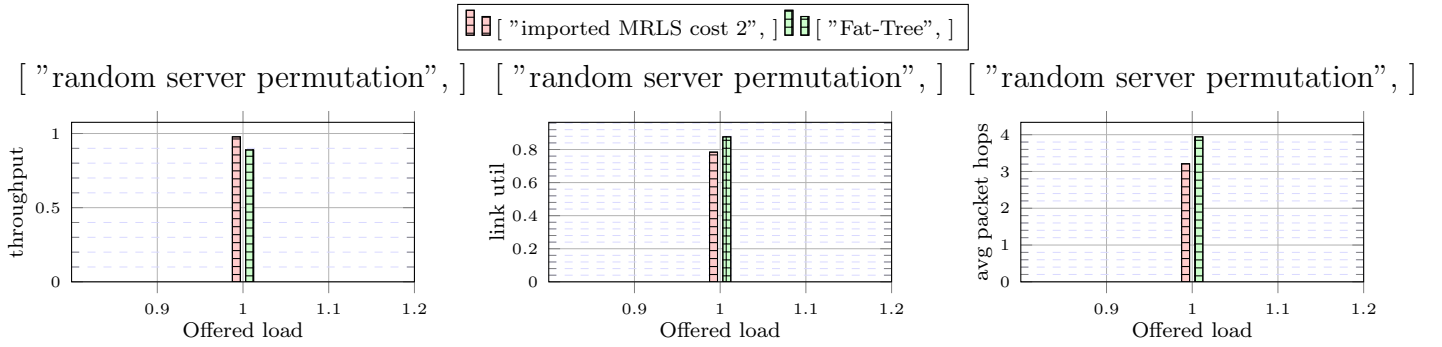


Figure 2: X: ["random server permutation",]

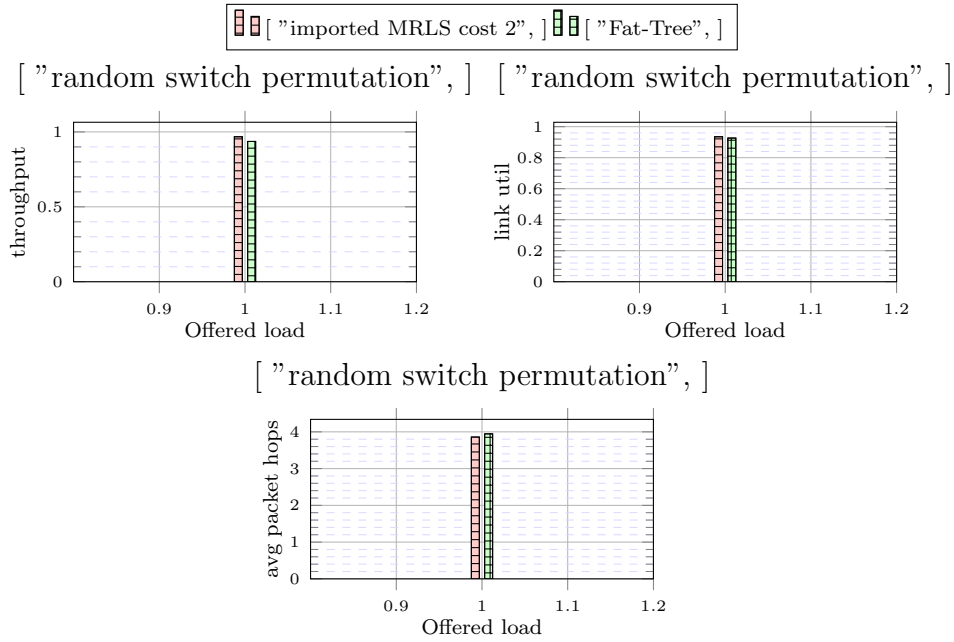


Figure 3: X: ["random switch permutation",]

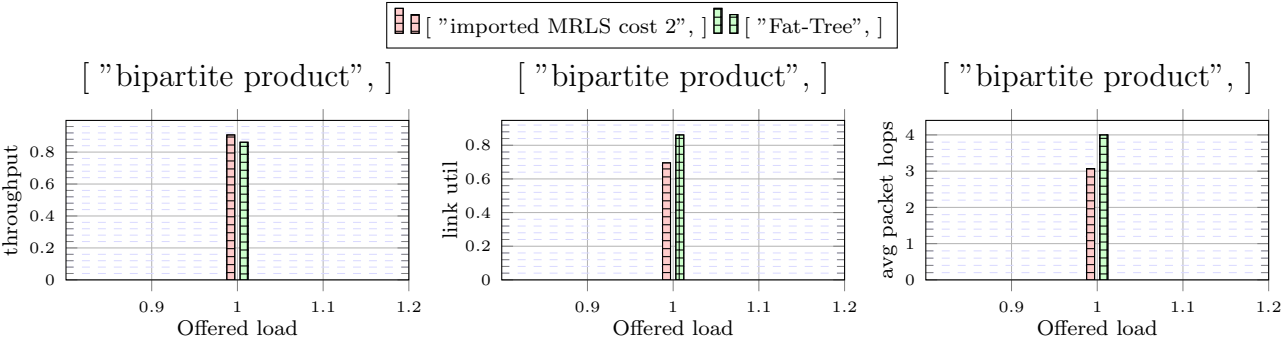


Figure 4: X: ["bipartite product",]

The following versions used in the simulations.

- heads/master-dirty-65e4f7db651d154c0abb3e3d0a691be7d6d92d82(0.6.3)
- heads/master-dirty-ce88a576295ce68f40736599a9e3e04f24f38801(0.6.3)
- heads/alex-dev-defe967c1db10e3b873fe074f274b6bdc9c42bb8(0.6.3)

```

Configuration{
  random_seed: ![ 42, 935, 128643 ],
  warmup: 20000,
  measured: 50000,
  general_frequency_divisor: 2,
  topology: topo![
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: ".../MRLS_972_24_648_36" },
        servers_per_leaf: 12,
        legend_name: "imported MRLS cost 2"},
      MultiStage{
        stages: [
          Fat{ bottom_factor: 18, top_factor: 18 },
          Fat{ bottom_factor: 36, top_factor: 18 }],
        servers_per_leaf: 18,
        legend_name: "Fat-Tree"}],
  traffic: HomogeneousTraffic{
    pattern: ![
      Uniform{ legend_name: "uniform" },
      RandomPermutation{ legend_name: "random server permutation" },
      CartesianTransform{
        sides: [ 18, 648 ],
        patterns: [
          Identity,
          RandomPermutation{ seed: 500 }],
        legend_name: "random switch permutation"},
      Product{
        block_size: 5832,
        block_pattern: Uniform,
        global_pattern: CartesianTransform{
          sides: [ 2 ],
          shift: [ 1 ],
          legend_name: ""},
        legend_name: "bipartite product"}],
    servers: 11664,
    load: ![ 0.5 ],
    message_size: 16},
  maximum_packet_size: 16,
  router: InputOutput{
    virtual_channels: topo![ 4, 2 ],
    virtual_channel_policies: topo![
      [
        MapHop{
          hop_to_policy: [
            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 2 ]},
            ArgumentVC{
              allowed: [ 2 ]},
            ArgumentVC{
              allowed: [ 3 ]},
            ArgumentVC{
              allowed: [ 3 ]}],
          VecLabel{
            label_vector: [ 0, 64, 80 ]},
            OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
            use_neighbour_space: true, aggregate: true },
            OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
            use_neighbour_space: true, aggregate: false },
            LowestLabel,
            EnforceFlowControl,
            Random],
          [ EnforceFlowControl, Random ]],
        allocator: Random,
        buffer_size: 128,
        bubble: false,
        flit_size: 16,
        intransit_priority: false,
        allow_request_busy_port: true,
        output_buffer_size: 64,
        crossbar_frequency_divisor: 1,
        crossbar_delay: 2},
    routing: topo![
      Polarized{ include_labels: true, strong: false, legend_name: "Polarized" },
      UpDown{ legend_name: "minimal routing" }],
    link_classes: [
      LinkClass{ delay: 2 },
      LinkClass{ delay: 2 },
      LinkClass{ delay: 2 }],
    launch_configurations: [
      Slurm{ job_pack_size: 1, time: "3-00:00:00" }]}

```